

MCV 4U Unit 7 -TEST PM

Name [REDACTED]

Date [REDACTED]

Total
31
/34

Knowledge

State whether the following is True or False. Provide reasons for any False answers.

1. T The angle between two vectors can be found using the cross product.

2. F The dot product of two perpendicular vectors is $\vec{0}$

Scalar, not vector

3. F If $\vec{a} \cdot \vec{b} = 7$, then $\vec{b} \cdot \vec{a} = -7$

$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$, Commutative property

4. The result of the following cross product $(\vec{i} - 2\vec{j}) \times (\vec{i} + 2\vec{j})$ is:

Let $\vec{a} = (\vec{i} - 2\vec{j}) = (1, -2, 0)$

$\vec{b} = (\vec{i} + 2\vec{j}) = (1, 2, 0)$

$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -2 & 0 \\ 1 & 2 & 0 \end{vmatrix} = (0, 0, 4)$

5. If $\vec{u} = (3a, 2a, a + 20)$ and $\vec{v} = a\vec{i} - 3\vec{j} - \vec{k}$, find the value(s) of a if $\vec{u}$ and $\vec{v}$ are perpendicular.

$\vec{u} \cdot \vec{v} = 0$

$(3a, 2a, a+20) \cdot (a, -3, -1) = 0$

$3a^2 - 6a - a - 20 = 0$

$3a^2 - 7a - 20 = 0$

$(a-4)(3a+5) = 0$

$a = 4$
 $a = -\frac{5}{3}$

7

6. Find the vector projection of $\vec{v}$ onto $\vec{u}$ given: $\vec{u} = (3, 6, -2)$, $\vec{v} = (-4, 3, 8)$

$$\frac{\vec{v} \cdot \vec{u}}{|\vec{u}|} \left(\frac{\vec{u}}{|\vec{u}|} \right) = \frac{(-4, 3, 8) \cdot (3, 6, -2)}{\sqrt{3^2 + 6^2 + (-2)^2}} \left(\frac{(3, 6, -2)}{\sqrt{3^2 + 6^2 + (-2)^2}} \right)$$

$$= \frac{-12 + 18 - 16}{\sqrt{49}} \left(\frac{(3, 6, -2)}{\sqrt{49}} \right) = \frac{-10}{7} \left(\frac{(3, 6, -2)}{7} \right) = \left(-\frac{30}{49}, -\frac{60}{49}, \frac{20}{49} \right)$$

7. A parallelogram RSTU is defined by 4 points, R(1,1,3), S(3,-3,6), T(0,0,10) and U(-2,4,7)

a) Find angle S of this parallelogram.

b) Find the area of this parallelogram

2-d visualization



$$\vec{SR} = \vec{OR} - \vec{OS} = (1, 1, 3) - (3, -3, 6)$$

$$\vec{SR} = (-2, 4, -3)$$

$$\vec{ST} = (0, 0, 10) - (3, -3, 6)$$

$$\vec{ST} = (-3, 3, 4)$$

tail to tail,

$$\vec{SR} \cdot \vec{ST} = |\vec{SR}| |\vec{ST}| \cos \theta$$

$$\cos \theta = \frac{(-2, 4, -3) \cdot (-3, 3, 4)}{(\sqrt{2^2 + 4^2 + 3^2})(\sqrt{3^2 + 3^2 + 4^2})}$$

$$\cos \theta = \frac{6 + 12 - 12}{(\sqrt{29})(\sqrt{34})}$$

$$\theta = \cos^{-1} \left(\frac{6}{\sqrt{986}} \right)$$

$$\theta = 79^\circ$$

$$b) \text{ Area} = |\vec{SR} \times \vec{ST}|$$

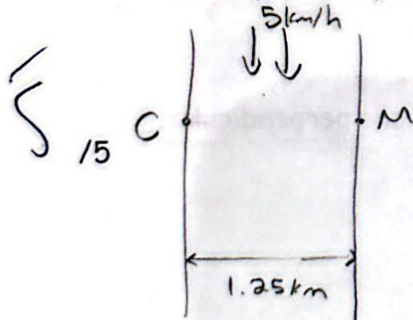
$$\vec{SR} + \vec{ST} = (25, 17, 6)$$

$$\begin{vmatrix} 4 & 3 \\ -3 & 4 \\ -2 & -3 \\ 4 & 3 \end{vmatrix}$$

$$|\vec{SR} \times \vec{ST}| = \sqrt{25^2 + 17^2 + 6^2} = \sqrt{950}$$

$$\text{Area} = 5\sqrt{38} \text{ u}^2$$

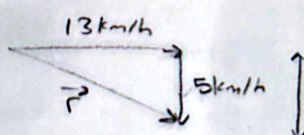
8. A canoeist wishes to cross a river 1.25km in width. The current flows at 5km/h and the canoeist can paddle at 13km/h in still water. If the marina is located directly across from the canoeist's starting point, how much longer (in seconds) would it take the canoeist to get to the marina by travelling in a direction perpendicular to the current versus changing her direction so that she ends up directly at the marina? (Hint: the canoeist never leaves the water)



let C be canoeist starting point
M be marina

dist = Speed x time

Route #1 (perpendicular)



$$\text{time} = \frac{\text{dist}}{\text{speed}} \quad \frac{1.25 \text{ km}}{13 \text{ km/h}} = 0.048 \text{ h}$$

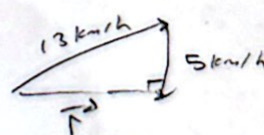
$$t_1 = \frac{1.25 \text{ km}}{13 \text{ km/h}}$$

$$t_1 = 0.04615 \text{ h}$$

$$t_{1a} = 546 \text{ s}$$

$$t_1 = 562 \text{ s}$$

Route #2, (directly across)



$$|\vec{r}| = \sqrt{13^2 - 5^2}$$

$$|\vec{r}| = 12 \text{ km/h}$$

$$t_2 = \frac{1.25 \text{ km}}{12 \text{ km/h}}$$

$$t_2 = 0.10417 \text{ h}$$

$$t_2 = 375 \text{ s}$$

Therefore it would take 187 seconds longer to traverse route #1.

9. Find a non-zero unit vector perpendicular to both $\vec{a} = \vec{i} + 2\vec{j} - \vec{k}$ and $\vec{b} = 2\vec{i} - \vec{k}$, without using cross product.

$$\vec{a} = (1, 2, -1)$$

$$\vec{b} = (2, 0, -1)$$

Let $\vec{c}$ be unit vector
 $\vec{c} = (x, y, z)$

$$\vec{a} \cdot \vec{c} = 0$$

$$x + 2y - z = 0$$

$$\vec{b} \cdot \vec{c} = 0$$

$$2x - z = 0$$

$$z = 2x$$

$$x + 2y - 2x = 0$$

$$x = 2y$$

$$(2y, y, 4y)$$

$$\frac{1}{|(2y, y, 4y)|} \cdot (2y, y, 4y)$$

pick a y

$$\sqrt{(2y)^2 + y^2 + 16y^2}$$

$$= \sqrt{4y^2 + y^2 + 16y^2}$$

$$= \sqrt{21y^2}$$

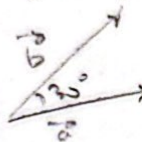
$$\frac{1}{\sqrt{21y^2}} \cdot (2y, y, 4y)$$

$$\frac{2y}{21y^2}, \frac{y}{21y^2}, \frac{4y}{21y^2}$$

$$= \left(\frac{2}{21y}, \frac{1}{21y}, \frac{4}{21y} \right)$$

10. Determine k algebraically, given two vectors and the angle between them.

$$\vec{a} = (2, 2\sqrt{3}) \quad \vec{b} = (0, k) \quad \theta = 30^\circ$$



$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$(2, 2\sqrt{3}) \cdot (0, k) = (\sqrt{2^2 + (2\sqrt{3})^2}) (\sqrt{k^2}) \cos 30^\circ$$

$$2\sqrt{3}k = (4)(k) \frac{\sqrt{3}}{2}$$

$$2\sqrt{3}k = 2\sqrt{3}k$$

$$\vec{a} \times \vec{b} = (0, 0, 2k)$$

$$|\vec{a} \times \vec{b}| = 2k$$

$$|\vec{a}| = 4$$

$$|\vec{b}| = k$$

$$|\vec{a} \times \vec{b}| = 4k \sin 30^\circ$$

$$2k = 2k$$

$$2k^2 - 2k = 0$$

$$2k(k-1) = 0$$

$$(k=0, 1)$$

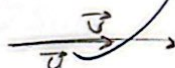
$$k > 0$$

~~It can be any real number.~~

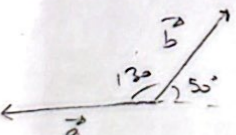
11. Illustrate the geometric vectors $\vec{u}$ and $\vec{v}$ if $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}|$. Justify.

According to dot product formula, $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$, and since $\cos(0^\circ) = 1$,

the angle between those two vectors must be 0° .

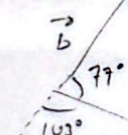


12. Determine the exact value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c}$ if $|\vec{a}| = 7$, $|\vec{b}| = 3$, $|\vec{c}| = 5$



$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos 130^\circ$$

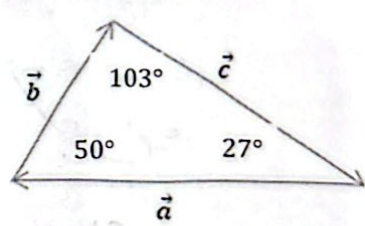
$$= (7)(3) \cos 130^\circ$$



$$\vec{b} \cdot \vec{c} = |\vec{b}| |\vec{c}| \cos 77^\circ$$

$$= (3)(5) \cos 77^\circ$$

$$\text{Therefore, } \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} = 21 \cos 130^\circ + 15 \cos 77^\circ$$



13. Given the points $X(1, -2, 1)$ and $Y(0, 3, -2)$, find a vector collinear with $\overrightarrow{XY}$ and having a magnitude of 5.

3
13

$$\begin{aligned}\overrightarrow{XY} &= \overrightarrow{OY} - \overrightarrow{OX} \\ &= (0, 3, -2) - (1, -2, 1) \\ &= (-1, 5, -3)\end{aligned}$$

Let $\vec{a}$ be unit vector
(collinear)

$$\begin{aligned}\overrightarrow{XY} &= k\vec{a} \\ \vec{a} &= (-k, 5k, -3k)\end{aligned}$$

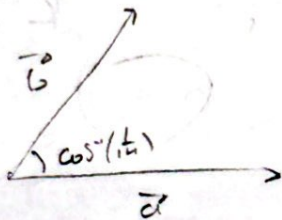
$$\begin{aligned}|\vec{a}| &= 5 \\ \sqrt{(-k)^2 + (5k)^2 + (-3k)^2} &= 5 \\ k^2 + 25k^2 + 9k^2 &= 25 \\ 35k^2 &= 25 \\ k^2 &= \frac{5}{7} \\ k &= \sqrt{\frac{5}{7}}\end{aligned}$$

$$\vec{a} = (-\sqrt{\frac{5}{7}}, 5\sqrt{\frac{5}{7}}, -3\sqrt{\frac{5}{7}})$$

$$\begin{aligned}\text{check} \\ |\vec{a}| &= \sqrt{\frac{5}{7} + 25\frac{5}{7} + 9\frac{5}{7}} \\ &= \sqrt{25} \\ &= 5\end{aligned}$$

14. The angle between the vectors $\vec{a} = (6, -2, 3)$ and $\vec{b} = (1, p, -2)$ is $\cos^{-1}(\frac{1}{14})$. Determine the value of p .

$$\cos^{-1}(\frac{1}{14}) = 85.9^\circ$$



$$\cos(\cos^{-1}(\frac{1}{14})) = \frac{1}{14}$$

~~$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$$~~

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$$

$$(6, -2, 3) \cdot (1, p, -2) = (\sqrt{6^2 + (-2)^2 + 3^2})(\sqrt{1^2 + p^2 + (-2)^2})(\frac{1}{14})$$

$$6 - 2p - 6 = 7(\frac{1}{14})(\sqrt{5+p^2})$$

$$-2p = \frac{1}{2}(\sqrt{5+p^2})$$

$$(-4p)^2 = (\sqrt{5+p^2})^2$$

$$16p^2 = 5 + p^2$$

$$15p^2 = 5$$

$$p^2 = \frac{1}{3}$$

$$p = \pm \frac{1}{\sqrt{3}}$$

can't be + ! gives $\frac{1}{14}$

6.5